

EFFECTS OF SOCIAL CUES ON SOCIAL SIGNALS IN HUMAN-ROBOT INTERACTION DURING A HALLWAY NAVIGATION TASK

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This study investigated how humans interact socially with robots. Participants engaged in a hallway navigation task with a robot. Throughout twelve trials, the display on the robot and its proxemics behavior was varied while participants were tasked with first, reacting to the robot's actions and second, interpreting its behavior. Results indicated that proxemic behavior and robotic display characteristics influence the degree to which individuals perceive the robot as socially present, with more human-like displays and assertive robotic behaviors resulting in greater assessments of social presence. When examined in isolation, repeated interactions over time do not appear to affect the perception of a socially present robot under these particular circumstances. Results are discussed in the context of how social signals theory inform research in human-robot interaction.

INTRODUCTION

As robots become more entrenched in situations with underlying social contexts, it is crucial for researchers in cognitive engineering to understand how to instantiate the necessary social skills that will allow them to fill the roles of teammates (Goodrich & Schultz, 2007; Phillips, Ososky, Grove, & Jentsch, 2011; Warta, Kapalo, Best, & Fiore, 2016). However, modern robots still face many challenges in both verbal and non-verbal communication during human-robot interaction (HRI). Extant research suggests that a contributing factor for successful HRI is the ability for a robot to display the most appropriate social cues during a given social scenario. Research in cognitive engineering advances our understanding of such processes by investigating the expression of robotic social cues and their influence on perceived social signals and how this supports advances in designing more effective social robots (Fiore et al., 2013; Hegel et al., 2011; Saulnier, Sharlin, & Greenberg, 2011; Warta, 2015; Wiltshire et al., 2015a).

In this paper we describe an interdisciplinary study that integrates social and cognitive psychology with methods and issues from robotics and computer science. We examine the relationship between social cues and social signals, a process we suggest drives mental state attribution between social agents. We review relevant research on proxemics and how it influences perception of social presence. We then detail our approach as well as the implications of our work in understanding how robotic social cues influence human perception of social signals to support cognitive engineering research on robotic design for improved HRI.

Social Cues and Social Signals

Social signal processing (SSP; Vinciarelli et al., 2012) is an interdisciplinary field linking technology with theory from social cognition. SSP draws from computer science and engineering to create a taxonomy of technologies to study how social behavior can be tracked and conveyed through technology. SSP has been used to ground research exploring the social elements of HRI (Fiore et al., 2013; Warta et al., 2016; Wiltshire et al., 2015a; 2015b; Wiltshire et al., 2013). We suggest this framework provides a foundation for understanding the relationship between robot behavior and human per-

ceptions of that robot through social cues and signals. Accordingly, an individual's social interactions are influenced by social cues, which drive social judgments (Wiltshire et al., 2015b; Wiltshire et al., 2014). *Social cues* include behavioral components whereas *physical cues* include features of physical appearance or aspects of the environment, such as the human-likeness or machine-likeness of a robot and the proxemic distance between interactors. *Behavioral cues* refer to non-verbal movements and gestures using the body and face as well as verbal communications (Fiore et al., 2013).

Through a combination of social cues, an individual can convey what becomes interpreted as a *social signal* representing emotions, intentions, or personality. For example, when two humans interact, one could provide a cue, such as a nod, to convey understanding another during a conversation. A robot could relay a social signal indicating its intention to yield to a human interactor's movements through the physical cue of slow speed. It is within this context that we explore what are the appropriate social cues for a robot to display.

Proxemics

Involving both distance and behavior, *proxemics* refers to the human use of space (Hall, 1963). The concept of proxemics has become increasingly important to study in the context of HRI as robots become more common and mobile. Indeed, previous research has shown that factors like age, personality, familiarity with robots, and gender all influence proxemic behavior in HRI (Rios-Martinez et al., 2014; Takayama & Pantofaru, 2009). Humans adjust proxemics when perceiving a robot as a social agent, where adjustments meant respecting the robot's personal space and not entering it without the robot's agreement (Straub, 2016). However, it is unclear whether a robot's cues can be perceived as social and whether social signals can be successfully conveyed in HRI if the robot is not perceived as a social agent (Warta et al., 2016; Wiltshire et al., 2013).

Social Presence

A social agent is considered socially present when it is perceived as capable of exhibiting mental states (Harms & Biocca, 2004) and studies show the influence of social presence on social interaction (Kennedy, Baxter, & Belpaeme,

2015; Pereira, Martinho, Leite, & Paiva, 2008). In particular, robotic agents perceived to possess greater social presence also have greater social influence (Thellman, Silvervarg, Gulz, & Ziemke, 2016). Social influence is characterized by the ability to affect behavior, which facilitates successful interactions. Research shows that individuals automatically lower their expectations of social presence when interacting with robots (Edwards et al., 2016; Spence et al., 2014). However, perceived social presence of robots can increase over repeated interactions (Fiore et al., 2013).

The Current Study

In this paper, we focus on human interaction with a physically embodied robot. Our goal was to build upon previous research studying how humans perceive and interpret robots with which they interact and their behavioral responses as a function of differing features of the robot (Fiore et al., 2013; Wiltshire et al., 2015; Wiltshire et al., 2013). We investigated how humans not only perceive, but also respond behaviorally to other agents and how their perception of, and reaction to, robots might differ under varying circumstances. We examined the understanding of intentions between humans and robots through the interpretation of social cues (e.g., assertive versus passive behavior) and explored how humans perceive robots when varying social cues as human-like or artificial (e.g., human-like, iconic, or symbolic display type).

This study is a replication and extension of our previous work investigating proxemic cues and mental state attribution (Fiore et al., 2013; Wiltshire et al., 2015a; Wiltshire et al., 2013). We developed a custom robot platform consisting of a mobile platform with a tablet affixed to the top to provide communication of social cues (see Figure 1). Participants interacted with the robot in a hallway navigation scenario. The robot presented one of three types of images conveying the manipulated social cues. Over the duration of the experiment, participants provided an immediate behavioral reaction in response to the interacting robot and answered questions relevant to the interaction.

Hypotheses

Our hypotheses are grounded in the two manipulated variables in this study (i.e., robot display type and proxemic behavior) and the dependent variable (i.e., social presence) evolving as a function of repeated exposure to the robot over twelve trials. We hypothesize that perceived social presence will be affected by: (a) variations in the robotic display type, (b) proxemic behavior, and (c) measurement trial.

Hypothesis 1. Variations in the robotic display type (i.e., human-like, iconic, symbolic) will produce an effect on perceived social presence, as a function of the degree of human-likeness exhibited by the display. Thus, with increasing degrees of human-likeness, the more participants will perceive the robot as a socially present agent.

Hypothesis 2. Proxemic behavior of the robot will have an effect on perceived social presence. Participants will perceive a greater degree of social presence, the more the robot behaves in a way that yields to participant navigation behavior (i.e., passive behavior).

Hypothesis 3. Perceived social presence will change over time, between measurement trials, from the first interaction with the robot until the last interaction with the robot.

METHOD

Participants

Participants were students from a large Southeastern university ($N = 102$; 54 females, 48 males). Ages ranged from 18 to 35 ($M_{age} = 19.79$). Students were compensated with course credit for their participation in the study.

Materials

The robotic platform was approximately 43" tall and consisted of a tablet mounted to a four-wheeled platform (see Figure 1). Included in this, affixed to the top of the robot, was a 12.3" Windows Surface Pro tablet that provided a display for the communication of social cues. The robotic platform was equipped with a Polhemus G4 Tracking system, which includes wireless six degrees-of-freedom trackers for position and orientation of the robot and participants with centimeter accuracy. Participants wore a tracker that the robot control software used to detect presence in real-time. For the experiment, we modified the Robotics Collaborative Technology Alliance's Multimodal Interface (MMI; Barber et al., 2015; Childers et al., 2015). This included a custom C++ programming language module to capture tracking data and operate the



robot to autonomously navigate the hallway such that it navigated towards the hallway exit point, either waiting for, or cutting in front of, the participant. The MMI also recorded timing and tracking data from the experiment in addition to when scenario events occurred in CSV log files. As such, this was not a traditional "Wizard of Oz" study. Rather, the robot was programmed to behave autonomously, not only depending on the condition, but also in response to the participant's own behavior. This was done to increase the ecological validity of the study as it made it more inline with actual autonomous interactions with robots.

Figure 1. Robotic platform.

To measure social presence, we used subscales of the Networked Minds Social Presence Inventory (NMSPI; Harms & Biocca, 2004). The subscales were: social co-presence (CP), attentional allocation (AA), perceived message understanding (PMU), and perceived behavioral interdependence (PBI). The survey was also modified such that instances of "my partner" were changed to "the robot" and the word "thoughts" in items

13 and 14 were changed to “intentions”. Responses were recorded on a 5-point Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*).

A rectangular hallway was constructed for this experiment in a large laboratory space. The main hallway measured 5 ft. by 29 ft., with a small 5 ft. by 4 ft. hallway alcove intersecting at the midpoint on one side (see Figure 2).

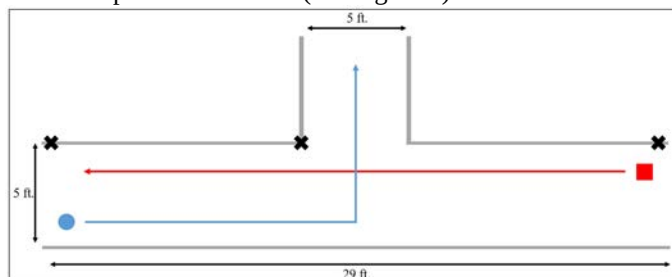


Figure 2. Hallway layout. The red square represents the participant starting point and the blue circle represents the robot starting point. The red and blue lines denote the paths to the trial end point for the participant and robot.

Design

The experiment was a 3 x (2 x 2) mixed model design. Display type was the three level between-factor (human-like, iconic, symbolic) crossed with two within-factor variables: proxemic behavior (assertive, passive) and measurement trial for the first and second measurement in a given proxemic condition. Participants were randomly assigned to one of the display type conditions and completed 12 trials in which they interacted with the robot during a hallway navigation task; six trials for the Assertive condition and six for the Passive condition. Proxemic conditions were counterbalanced between participants. Responses to the subjective measures were collected after the first and sixth trial for both of the proxemic conditions, resulting in a total of four measurement trials.

	Assertive	Passive
Symbolic		
Iconic		
Human-like		

Figure 3. Display types by proxemic condition.

Independent Variables (IVs). The IVs were the robotic social cues manipulated during each trial: (a) display type and (b) proxemic behavior. The *display type* was operationalized in terms of the graphic presented on the tablet and had three levels: human-like, iconic, and symbolic. The graphic displayed for these three levels varied as a function of the proxemic behavior condition (see Figure 3). *Proxemic behavior* was operationalized through control of the robot’s speed and pathing behavior at two levels: Assertive and Passive. During *Assertive* behavior, the robot sped up and modified its path to appear to “cut off” participants at the corner of the hallway alcove intersection, resulting in the robot passing in front of

the participant (see Figure 2). During *Passive* behavior, the robot traveled at a slower speed and modified its path to “give way”, providing the participant with space to pass the robot before it turned into the alcove.

Dependent Variables (DVs). The DVs in this study were: (a) participant reactions to the robot’s behavior and (b) responses to the subjective measures. These responses included assessments of: state-trait anxiety, mental state attributions, social presence, interpersonal attributions, and the robot’s interaction behaviors. However, due to space limitations, the current paper only reports results relating to social presence.

Procedure

Participants were brought into the research site and provided with an informed consent document to review before the experiment began. Once consent to participate was documented, participants completed a measure of state-trait anxiety and were introduced to the robot. After learning about the robot, participants were asked to perform a hallway navigation task, interacting with the robot in the constructed hallway. Once participants completed the first trial, they were asked to complete a series of subjective measures, which included the NMSPI. These measures were also completed following trials six, seven, and twelve. Lastly, participants were asked to complete a demographics questionnaire.

RESULTS

Social Presence

We present a series of analyses examining the effects of the IVs on the overall measure of social presence. A 3 (display type) x 2 (proxemic behavior) x 2 (measurement time) repeated-measures ANOVA was run. The between-subjects variable was display type (human-like, iconic, and symbolic) and the within-subjects variables were proxemic behavior (assertive and passive) and measurement time (first and second measurement in a given proxemic condition) with the overall scores on the NMSPI as the DV. All post-hoc analyses utilized Bonferroni corrections to adjust for multiple comparisons. The reliability for the NMSPI was approximately $\alpha = 0.86$.

First, there was a significant main effect of proxemic behavior on overall social presence, $F(1,99) = 30.42, p < .0001, \eta_p^2 = .24$, observed power = 1.00. Participants reported a higher level of social presence during the assertive condition ($M = 2.48$) than the passive condition ($M = 2.23$).

Second, there was a significant main effect of display type on overall social presence, $F(2,99) = 3.44, p = .036, \eta_p^2 = .07$, observed power = .63. Post-hoc analyses revealed that participants reported a higher level of social presence when in the human-like display condition ($M = 2.51$) compared to the iconic condition ($M = 2.25$). The reported levels of social presence were similar between all other conditions.

Third, there was a significant interaction between proxemic behavior and measurement time, $F(1,99) = 5.79, p = .018, \eta_p^2 = .06$, observed power = .66. Follow-up pairwise comparisons were conducted to better understand the interaction between the robot’s proxemic behavior and measurement time. First, there was a difference in social presence between proxemic behavior conditions. Specifically, there was a significant

difference in the levels of social presence reported during measurement time one, with higher levels observed in the assertive condition ($M = 2.45$, $SE = .06$) when compared to the passive condition ($M = 2.28$, $SE = .05$, $p = .006$). Similarly, during measurement time two, significantly higher levels of social presence were observed in the assertive condition ($M = 2.52$, $SE = .06$) when compared to the passive condition ($M = 2.18$, $SE = .05$, $p < .0001$). Second, there was a difference in levels of social presence between measurement times. There were significantly higher scores of social presence reported for the passive condition between measurement time one ($M = 2.28$, $SE = .05$) and two ($M = 2.18$, $SE = .05$, $p = .041$).

Lastly, there was a significant interaction between proxemic behavior, measurement time, and display type, $F(2,99) = 3.85$, $p = .025$, $\eta_p^2 = .07$, observed power = .69. Follow-up pairwise comparisons were conducted to better understand the interaction between the robot's proxemic behavior, measurement time, and display (see Figure 4). Levels of social presence for each proxemic behavior condition differed during measurement times and between display types. The levels of social presence reported during T1 for the symbolic display in the assertive condition ($M = 2.53$, $SE = .10$) were significantly higher than in the passive condition ($M = 2.13$, $SE = .09$, $p < .0001$). Additionally, the levels of social presence reported in the assertive condition were significantly higher than in the passive condition during T2 for the human-like display ($M_{Assertive} = 2.65$, $SE = .10$; $M_{Passive} = 2.31$, $SE = .08$, $p < .0001$), the iconic display ($M_{Assertive} = 2.43$, $SE = .10$; $M_{Passive} = 2.10$, $SE = .08$, $p < .0001$), and the symbolic display ($M_{Assertive} = 2.46$, SE

urement time. Specifically, there was a significant difference in the levels of social presence reported in the passive condition during measurement time one, with higher levels of social presence reported for the human-like display ($M = 2.55$, $SE = .09$) when compared to the iconic ($M = 2.16$, $SE = .09$, $p = .005$) and symbolic displays ($M = 2.13$, $SE = .09$, $p = .002$).

DISCUSSION

We investigated the effects of display types, proxemic behavior, and repeated interactions on the perception of social presence during a hallway navigation task between a human and robot. In particular, social presence was measured to better understand the degree to which individuals perceived the robot as a social agent, as a function of which proxemic behaviors and display type it was exhibiting, and whether repeated interactions would alter this perception.

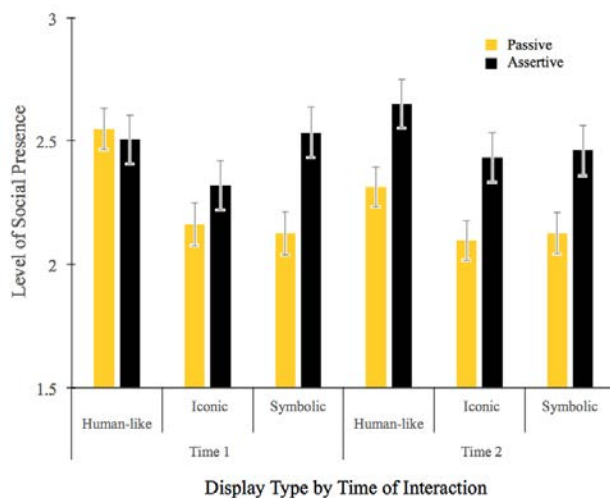
In support of *Hypothesis 1*, display type, particularly the human-like display, had a significant effect on social presence, as indicated by the main effect for display type and three-way interaction between proxemic behavior, measurement time, and display type. When display type is examined in isolation, it appears that social presence is higher in the human-like display when compared to the iconic display. However, during the first measurement trial when the robot was exhibiting passive behavior, social presence was higher for the human-like display when compared to all other display types. Overall, these results indicate that when it comes to displays, greater degrees of human-likeness and thus, more familiar features, support the perception of higher levels of social presence.

Contrary to *Hypothesis 2* and Wiltshire et al. (2013), passive proxemic behavior did not yield greater levels of social presence. The passive behavior settings were meant to support the perception of the robot as a social actor in which it behaved deferentially toward participants by yielding to their movements. It may be that social presence was perceived to be higher in the assertive condition because the robot appeared to have a mind of its own when it came close to running into participants or otherwise acted contrary to its programming and participants' expectations of a robot simply passing them in the hallway. In particular, this may have given participants the perception that the robot was a social agent.

Contrary to *Hypothesis 3* and Wiltshire et al. (2013), time did not have a significant effect on perceptions of social presence when examined in isolation. However, social presence was significantly higher during the first measurement of the passive condition. Wiltshire et al. (2013) credits behavioral consistency as a fundamental component to the perception of a socially present robot. Thus, these results suggest that no significant change in social presence over time may be evidence of participants' perception of the robot behaving inconsistently across all 12 trials.

CONCLUSION

In this study, we investigated how humans not only perceive, but also respond behaviorally, to other agents and how their perception of, and reaction to, robots might differ under varying circumstances across repeated interactions. Using an



$= .10$; $M_{Passive} = 2.13$, $SE = .08$, $p < .0001$).

Figure 4. Interaction between proxemic behavior, measurement time, and display type.

Second, the levels of social presence reported during each measurement time differed significantly in the human-like display condition. Accordingly, the levels of social presence reported in the passive condition for the human-like display were significantly higher during measurement time one ($M = 2.55$, $SE = .09$) than measurement time two ($M = 2.31$, $SE = .08$, $p = .005$).

Last, the levels of social presence reported for each display type differed as a function of proxemic behavior and meas-

interdisciplinary approach for cognitive engineering research, we integrated concepts and methods for the social and computational sciences with robotics to study how social cues and social signals relate to mental state attributions made by humans when interacting with robots.

While the task we chose for the HRI was confined to hallway navigation, this represents our main limitation. Accordingly, we acknowledge that our results may not generalize to other contexts that may be more socially interactive and involve a different robotic platform. However, the process of navigation is an interaction that may become more common between humans and robots in the future. As such, this study provides more information on the issues presented in Wiltshire et al. (2013) and demonstrates the need for further research addressing social cues and signals in HRI.

In sum, this experiment set out to study how social cues expressed by a robot affect human perception of social presence. To meet this objective, we ground this work within the framework of social signals processing and focused on the social qualities of a robot platform. This research has implications for understanding how social cognitive processes, derived from research on human-human interactions, do or do not translate to human-robot interactions. The question for research in cognitive engineering is to understand how human social cognitive mechanisms elicit the perception of machines as social agents, and how that alters human-robot interactions. This research can inform, not only the development of robots perceived as socially present, but also how such features influence more general social-cognitive processes such as they occur in teamwork.

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